

Allegro Current Sensor ICs Providing Green Energy Solutions in Photovoltaic (PV) Inverters



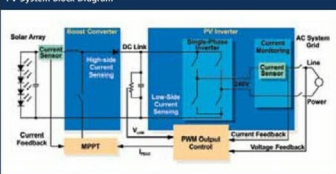
Allegro's family of fully integrated current sensor and linear ICs offer high accuracy sensing solutions for grid tied PV solar systems. Allegro's surface mount packages provide small form factor, fully integrated current sensor ICs with galvanic isolation. These ICs can sense 5 to 50 A of peak current and are rated to 3000 VRMS isolation. The series resistance of the package primary conductor is nominally 1 mOhm and provides for highly efficient current measurement in PV applications.

The Allegro ACS770 family of parts delivers an even lower 100 uOhm series resistance and is capable of sensing continuous current from 50 to 200 A and are rated up to 4800 VRMS isolation. Allegro's A1363 family of devices is available in a 1 mm SIP package and capable of sensing current greater than 200 A.

Allegro Advantages for Solar Inverters

- High accuracy ZERO Amp output voltage and device sensitivity achieved with Allegro factory programming
- Capable of sensing AC current and DC current
- Cost-effective solution compared to closed loop sensor technology
- 4800 VRMS isolation
- High temperature operation up to 150°C
- 100 uOhm and <1 mOhm internal conductor resistance for low power dissipation
- Small factor, low profile packages compared to current transformers

PV System Block Diagram



A1363	ACS770	ACS710
Sensed Current > 200 A	Sensed Current 50 to 200 A	Sensed Current 5 to 50 A

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